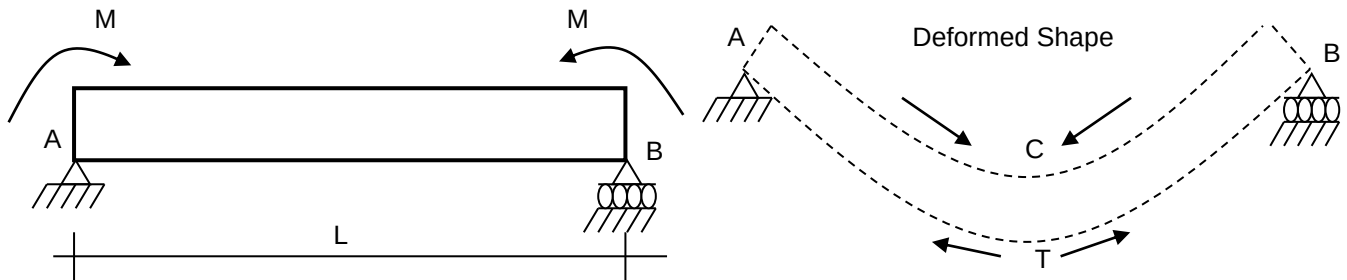


FCC3 232 CHAP 1 FLEXURAL STRESS IN BEAMS

1.1 INTRODUCTION

Beams are structural members which are predominantly subjected to **flexure**. Consider the simply supported beam shown below, which is subjected to two equal end moments **M**



The beam will carry the applied bending moments by changing its geometry from that of a **straight line** to that of a **curve**. This phenomenon is referred as **flexing**. **Flexure** is the technical term for **bending**. **Flexure** induces a **tension zone** in the bottom fibers and a **compression zone** in the top fibers, with a **neutral axis** dividing the two zones.

At any given cross section, the beam will be subjected to **flexural stresses only** (since there is no shear or axial stresses). The beam is therefore said to be subjected to **pure flexure**. In this chapter, we shall develop **mathematical equations** for the **flexural stresses** induced in a beam section due to pure flexure.

1.2 EULER'S THEORY OF PURE FLEXURE

[No shear forces, No axial force]

The following **assumptions** are made in the **theory of pure flexure** which was pioneered by Leonhard **Euler** and Danielle **Bernoulli** [1750]: -

1. The beam material is **homogeneous, isotropic, linearly elastic**, has the same value of **Young's Modulus** both in tension and in compression, and is not stressed beyond the **elastic limit**.
2. The beam section is **prismatic, initially straight**, and all longitudinal fibers bend in **circular arcs** with a **common center of curvature** and a **radius of curvature** which is large compared to the beam cross section dimensions.
3. The induced flexural stresses are **uniform** across the **width of the section**.
4. **Plane transverse sections remain plane** after bending and perpendicular to the Neutral Axis.

1.3 EULER'S EQUATIONS

Consider a small element of the beam, whose curvature has been exaggerated for clarity. Let R be the radius of curvature (measured from the centre of curvature O , to the neutral axis), and $d\theta$ be the change in slope between the ends of the considered element.

Consider the fibers AA distance y from the NA.

Original length of fibers AA = $R d\theta$

Final length of fibers AA = $(R + y) d\theta$

Flexural strain = $[(R + y) d\theta - R d\theta] / R d\theta$

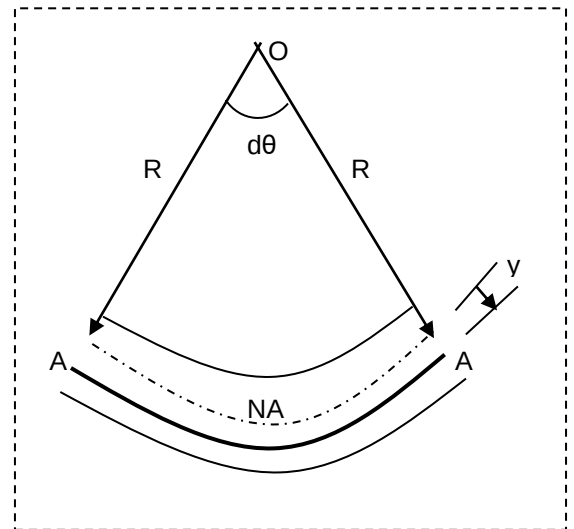
Thus $\epsilon = y/R$

The flexural stress is given by $f = E\epsilon$

Thus $f = Ey/R$

Hence we can write

$f/y = E/R$ ----- Eqn 1



Longitudinal force in fibers AA is given by $dF = f dA = (E/R) y dA$

The total longitudinal force is obtained by integrating dF over the whole cross section from bottom to top fibers. In the case of pure bending, the total longitudinal force must be zero (since there is no axial force) $F = \int_A dF = \int_A (E/R) y dA = (E/R) \int_A y dA = 0$

Since $E/R \neq 0$, then $\int_A y dA = 0$. And since $\int_A y dA = 1^{\text{st}}$ moment of area, then it means that the **Neutral Axis coincides with the centroidal x-axis of the cross section**

Equilibrium of moments requires that the applied moment be equal and opposite to the moment of resistance of the section, thus

$$M = \int_A y f dA = \int_A y (E/R) y dA = (E/R) \int_A y^2 dA$$

But $\int_A y^2 dA = I$, the second moment of area of the section (aka the moment of inertia)

Thus $M = (E/R) I$ Hence we can write **$M/I = E/R$ ----- Eqn 2**

Substitute Eqn 1 into Eqn 2 **$M/I = f/y$** and rearrange **$f = My/I$ ----- Eqn 3**

Eqn 3 (called **Euler's Eqn**) is used to determine the **flexural stresses** in a beam section.

Eqns 1 and 2 above can be combined to form the classical **Euler-Bernoulli** equation which is expressed as **$f/y = E/R = M/I$ ----- Eqn 4**

1.4 FLEXURAL STRESS DISTRIBUTION

In Eqn 3, the bending moment **M** is applied about the **x-axis**, and the 2nd moment of area **I** is calculated about the **x-axis**. Eqn 3 can therefore be written as $f_y = (M_x/I_x) y$. The ratio **M_x/I_x** is a constant at any given cross section. This means that the **flexural stress is directly proportional to the distance of the fibers from the Neutral Axis, NA**.

The fibers above the **NA** will be subjected to compression, while the fibers below the NA will be subjected to tension.

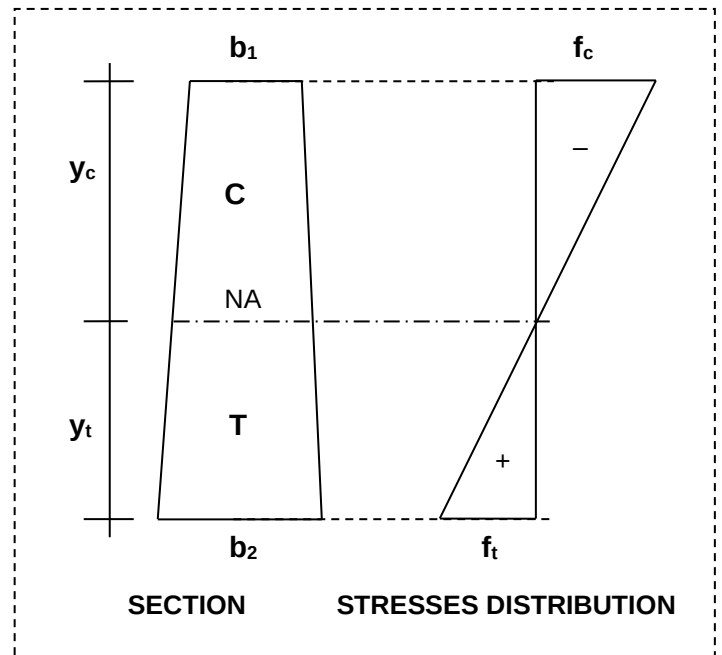
By sign convention, the **tensile stresses** are designated **positive**, while the **compressive stresses** are designated **negative**.

Eqn 3 may be written as $f_y = M_x/(I_x/y)$.

The quantity **I_x/y** is a constant at any level of fibers. At the extreme bottom fibers, the quantity **I_x/y_t** is called the **section modulus in tension**, denoted by **Z_t**.

Similarly, the quantity **I_x/y_c** is called the **section modulus in compression**, denoted by **Z_c**.

If the section is symmetric about the x-axis, then **Z_t = Z_c = Z**, say.



The **maximum tensile** and **maximum compressive stresses** are therefore given by

$$f_t = M/Z_t \quad \text{and} \quad f_c = M/Z_c \quad \text{-----} \quad \text{Eqns 5}$$

In simple beam design, the applied moment **M** is given and the allowable stresses **f_t** and **f_c** are prescribed. The design work involves choosing and specifying a section whose **Z_t** and **Z_c** are such that the prescribed allowable stresses will not be exceeded. Eqns 5 are therefore called **beam design equations**.

If the bending moment **M** is applied about the **y-axis**, then the 2nd moment of area **I** is calculated about the **y-axis** and Eqn 3 is written as $f_x = (M_y/I_y) x$. ----- Eqn 3*

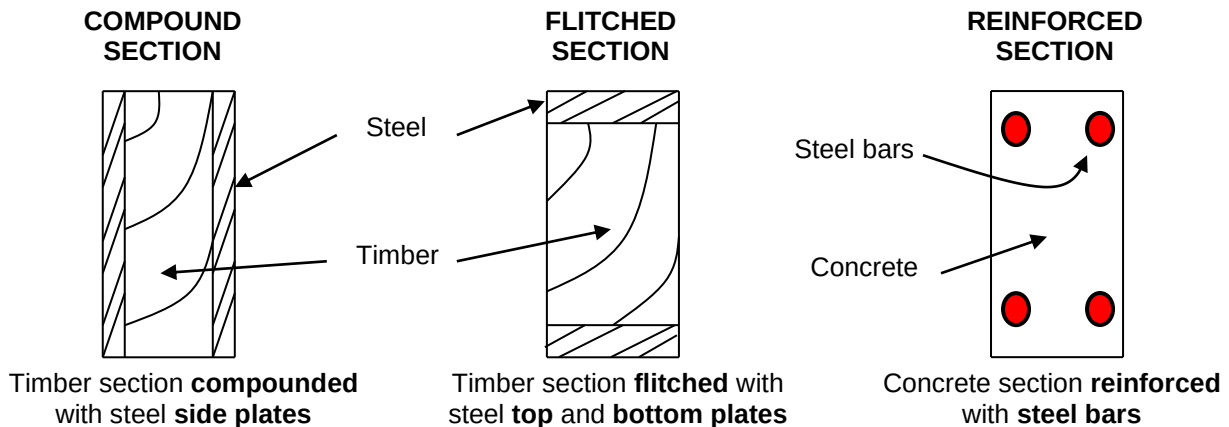
In practice, the effects of shear and axial stresses are negligible compared to the effects of flexural stress. Eqns 3, 5 and 3* can therefore be applied to any beam.

1.5 COMPOSITE SECTIONS

A weaker material may be combined with a stronger material in order to make the beam more structurally efficient. A composite section is a combination of two materials in the same section. There are three types of composite sections as illustrated below.

Compound and **flitched** sections are within the scope of **FCC3 232**.

[Reinforced concrete section shall be dealt with in FCE 332]



In a **compound section**, the weaker material is strengthened by **compounding plates** made of a stronger material. The compounding plates have the **same depth** as the weaker material, but the **breadth is smaller**. The compounding plate may be placed on **one side** or on **both sides** or **through the center** of the weaker material.

In a **flitched section**, the weaker material is strengthened by **flitching plates** made of a stronger material. The flitching plates have the **same breadth** as the weaker material, but the **depth is smaller**. The flitching plate may be placed **at the top**, or **at the bottom**, or **both at the top and at the bottom** sides of the weaker material.

The analysis of stress distribution in composite sections is based on the **assumption** that the materials are **sufficiently bonded** along the interface so that they attain the same **curvature** when subjected to a bending moment.

1.6 EQUIVALENT 2nd MOMENT OF AREA

Consider the compound section shown below which is subjected to a bending moment **M** acting about the **x-axis**.

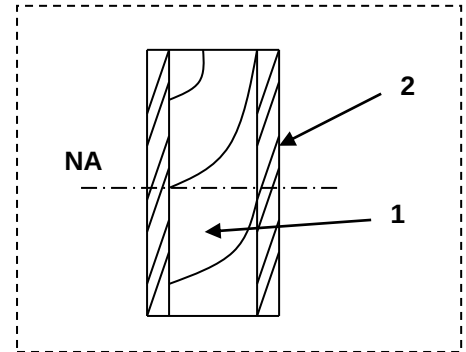
Equilibrium requires that $M = M_1 + M_2$

Where **M** is the applied bending moment

M_1 is the moment resisted by material 1

And M_2 is the moment resisted by material 2

Using Eqn 2: $M = E_1 I_1 / R_1 + E_2 I_2 / R_2$



Compatibility requires that $1/R_1 = 1/R_2 = 1/R$

Therefore $M = (E_1 I_1 + E_2 I_2) / R$

Constitutivity requires that $1/R = f_1 / E_1 y$

Therefore $M = (E_1 I_1 + E_2 I_2) (f_1 / E_1 y)$

The above equation can be written as $f_1 = My / [I_1 + (E_2 / E_1) I_2]$

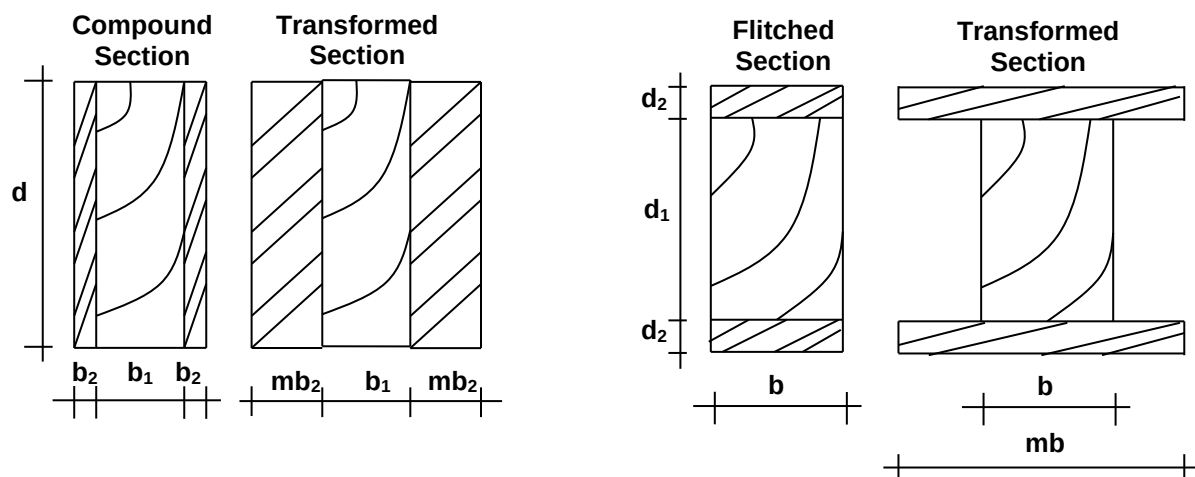
If we define $m = E_2 / E_1$ then $f_1 = My / [I_1 + m I_2]$ OR $f_1 = My / I_E$ ----- Eqn 6

If we express $1/R = f_2 / E_2 y$ then it can be shown that $f_2 = m f_1$ ----- Eqn 7

The quantity $m = E_2 / E_1$ is called the **modular ratio** (Note that $E_2 / E_1 \geq 1$)

The quantity $I_E = [I_1 + m I_2]$ is called the **equivalent 2nd moment of area**.

The **Equivalent 2nd Moment of Area** may be interpreted as illustrated below.

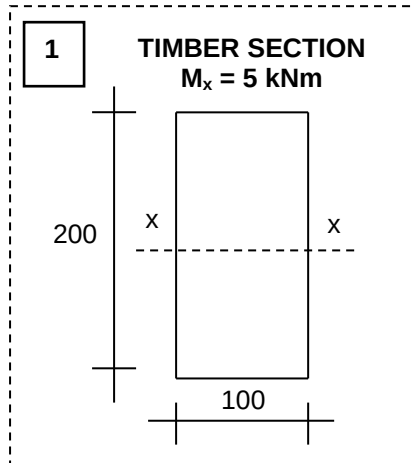


1.7 WORKED EXAMPLES

Determine the **tension** and **compression stresses** at the top and bottom in each section.

ALL dimensions are in mm. [Rectangle $I_x = bd^3/12$ Triangle $I_x = bh^3/36$]

$[E_s = 200 \text{ kN/mm}^2 \quad E_t = 10 \text{ kN/mm}^2]$



Solution

$$I_x = bd^3/12 = 100(200^3)/12 = 66.67 \times 10^6 \text{ mm}^4$$

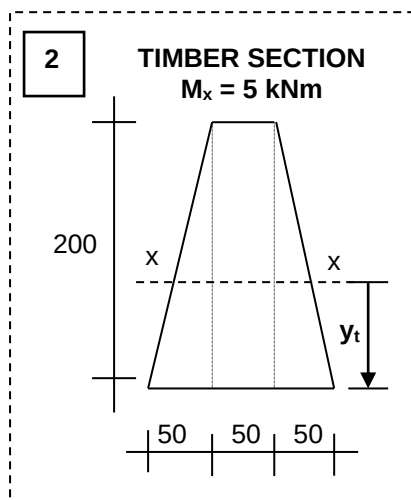
$$y_t = y_c = 100 \text{ mm}$$

$$Z_t = Z_c = I_x/y_t = 666.67 \times 10^3 \text{ mm}^3$$

Using Eqns 5

$$f_t = M_x/Z_t = 5 \times 10^6 \text{ Nmm} / (666.67 \times 10^3 \text{ mm}^3) = 7.5 \text{ N/mm}^2$$

$$f_c = M_x/Z_c = 5 \times 10^6 \text{ Nmm} / (666.67 \times 10^3 \text{ mm}^3) = 7.5 \text{ N/mm}^2$$



Solution $[A = 2 \times \frac{1}{2} \times 50 \times 200 + 50 \times 200 = 20 \times 10^3 \text{ mm}^2]$

$$y_t = [2 \times \frac{1}{2} \times 50 \times 200 \times \frac{1}{3} \times 200 + 50 \times 200 \times 100] / A = 83.3 \text{ mm}$$

$$I_x = 2[bh^3/36 + \frac{1}{2} \times 50 \times 200 \times (y_t - \frac{1}{3} \times 200)^2] + bd^3/12 + bd(100 - y_t)^2 = 61.11 \times 10^6 \text{ mm}^4$$

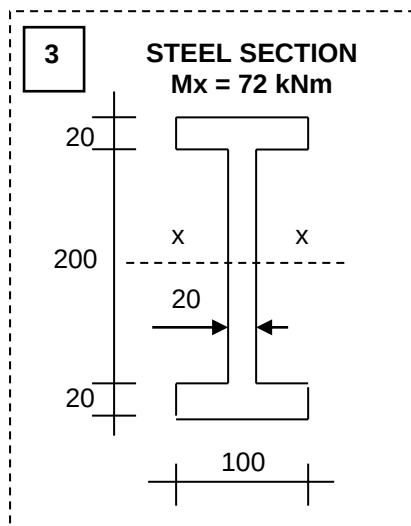
$$y_t = 83.3 \text{ mm} \quad y_c = 200 - y_t = 116.7 \text{ mm}$$

$$Z_t = I_x/y_t = 733.61 \times 10^3 \text{ mm}^3 \quad Z_c = I_x/y_c = 523.65 \times 10^3 \text{ mm}^3$$

Using Eqns 5

$$f_t = M_x/Z_t = 5 \times 10^6 \text{ Nmm} / (733.61 \times 10^3 \text{ mm}^3) = 6.8 \text{ N/mm}^2$$

$$f_c = M_x/Z_c = 5 \times 10^6 \text{ Nmm} / (523.65 \times 10^3 \text{ mm}^3) = 9.5 \text{ N/mm}^2$$



Solution

$$I_x = 100 \times 240^3/12 - 80 \times 200^3/12 = 61.87 \times 10^6 \text{ mm}^4$$

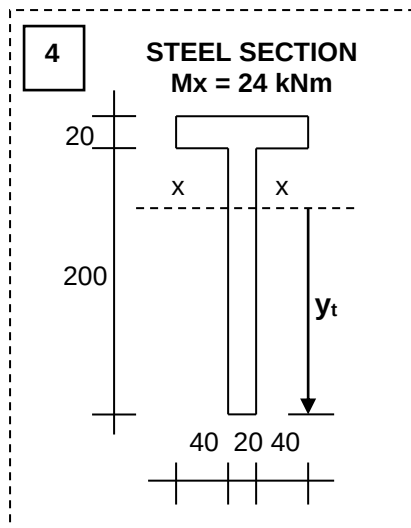
$$y_t = y_c = 120 \text{ mm}$$

$$Z_t = Z_c = I_x/y_t = 515.6 \times 10^3 \text{ mm}^3$$

Using Eqns 5

$$f_t = M_x/Z_t = 72 \times 10^6 / (515.6 \times 10^3) = 139.6 \text{ N/mm}^2$$

$$f_c = M_x/Z_c = 72 \times 10^6 / (515.6 \times 10^3) = 139.6 \text{ N/mm}^2$$



Solution $[A = 20 \times 200 + 100 \times 20 = 6.0 \times 10^3 \text{ mm}^2]$

$$y_t = [20 \times 200 \times 100 + 100 \times 20 \times 210] / A = 136.7 \text{ mm}$$

$$I_x = 20 \times 200^3 / 12 + 20 \times 200 \times (y_t - 100)^2 + 100 \times 20^3 / 12 + 100 \times 20 \times (210 - y_t)^2 = 29.53 \times 10^6 \text{ mm}^4$$

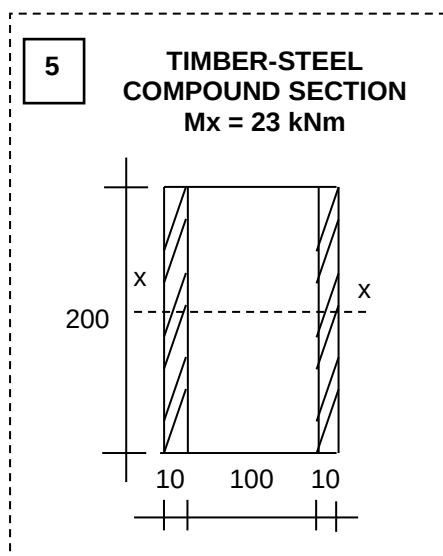
$$y_t = 136.7 \text{ mm} \quad y_c = 220 - y_t = 83.3 \text{ mm}$$

$$Z_t = I_x / y_t = 216.02 \times 10^3 \text{ mm}^3 \quad Z_c = I_x / y_c = 354.50 \times 10^3 \text{ mm}^3$$

Using Eqns 5

$$f_t = M_x / Z_t = 24 \times 10^6 / (216.02 \times 10^3) = \mathbf{111.1 \text{ N/mm}^2}$$

$$f_c = M_x / Z_c = 24 \times 10^6 / (354.50 \times 10^3) = \mathbf{67.7 \text{ N/mm}^2}$$



Solution [Let Timber = 1 and Steel = 2]

$$I_1 = 100 \times 200^3 / 12 = 66.67 \times 10^6 \text{ mm}^4$$

$$I_2 = 2 \times 10 \times 200^3 / 12 = 13.33 \times 10^6 \text{ mm}^4$$

$$\text{Modular ratio } m = E_s / E_t = 200 / 10 = 20$$

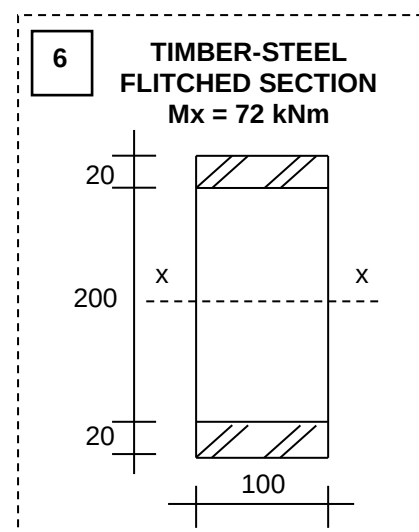
$$I_E = I_1 + m I_2 = 333.27 \times 10^6 \text{ mm}^4$$

Using Eqns 6 and 7

$$\text{Timber: } f_t = M_y / I_E = 23 \times 10^6 \times 100 / I_E = \mathbf{6.9 \text{ N/mm}^2}$$

$$\text{Steel: } f_t = m x M_y / I_E = 20 \times 23 \times 10^6 \times 100 / I_E = \mathbf{138.0 \text{ N/mm}^2}$$

[Symmetry: The stresses are the same on both sides]



Solution [Let Timber = 1 and Steel = 2]

$$I_1 = 100 \times 200^3 / 12 = 66.67 \times 10^6 \text{ mm}^4$$

$$I_2 = 100 \times 240^3 / 12 - 100 \times 200^3 / 12 = 48.53 \times 10^6 \text{ mm}^4$$

$$\text{Modular ratio } m = E_s / E_t = 200 / 10 = 20$$

$$I_E = I_1 + m I_2 = 1037.27 \times 10^6 \text{ mm}^4$$

Using Eqns 6 and 7

$$\text{Timber: } f_t = M_y / I_E = 72 \times 10^6 \times 100 / I_E = \mathbf{6.9 \text{ N/mm}^2}$$

$$\text{Steel: } f_t = m x M_y / I_E = 20 \times 72 \times 10^6 \times 120 / I_E = \mathbf{166.6 \text{ N/mm}^2}$$

[Symmetry: The stresses are the same on both sides]